

Sharpness of Irrationality after One Stabilization for Cubic Threefolds

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August 2026

Abstract

The irrationality of $X \times \mathbf{P}^1$ for every smooth complex cubic threefold X is sharp: two explicit smooth cubic threefolds over $\mathbf{Q}$ are rational after multiplication by $\mathbf{P}^2$. For each, the least m such that $X \times \mathbf{P}^m$ is rational is two, over both $\mathbf{Q}$ and $\mathbf{C}$. We prove more generally that if k has characteristic zero, S is a smooth quartic del Pezzo surface with $S(k) \neq \emptyset$, and $\text{Pic}(S_{\bar{k}})$ is a stably permutation Galois module, then $S \times \mathbf{A}^2$ is k -rational. We identify a rational quotient of the projective Cox model by a saturated rank-three subtorus; the residual two-dimensional torus is rational. Applying this theorem to the quartic del Pezzo fibrations of Tschinkel and Zhang, we prove that every member of both cubic series is rational after multiplication by $\mathbf{P}^2$, giving a uniform two-dimensional bound for their stable-rationality results. Consequently the stabilization level is two on explicit examples in cubic-threefold moduli and infinite at a very general point. We also obtain smooth projective nonrational fourfolds $Y/\mathbf{Q}$ such that $Y \times \mathbf{A}^1$ is rational.

Keywords. cubic threefold; quartic del Pezzo surface; stable rationality; universal torus; variety with one apparent double point; algebraic torus; rationality after affine-line stabilization.

1 Introduction and main results

Tschinkel and Zhang constructed explicit stably rational smooth cubic threefolds over $\mathbf{Q}$ [7, Theorem 1.3]. On the other hand, the one-stabilization theorem shows that every smooth complex cubic threefold remains irrational after multiplication by $\mathbf{P}^1$ [5, Theorem 1.1]. Thus at least two stabilizing variables are needed for the examples of Tschinkel and Zhang. We prove that this lower bound is attained by two of their examples.

For a projective variety V/k , put

$$\ell_k(V) = \min\{m \geq 0 : V \times \mathbf{A}^m \text{ is } k\text{-rational}\}.$$

If the set is empty, put $\ell_k(V) = \infty$. Since $\mathbf{A}^m$ and $\mathbf{P}^m$ are birational, either factor gives the same invariant. This is the level of stable rationality considered by Tschinkel and Zhang [7, p. 13]. Rationality is preserved by extension of the ground field, so $\ell_{\mathbf{Q}}(V) \geq \ell_{\mathbf{C}}(V_{\mathbf{C}})$ for $V/\mathbf{Q}$.

Consider the smooth cubic threefolds

$$X_1 : (x_4 - 2x_3)x_1^2 + 3(x_4 + 2x_3)x_2^2 + 3x_3^2x_4 - x_4^3 + x_5^3 = 0, \quad (1)$$

$$X_3 : x_4(x_1^2 + 2x_1x_2) + x_3(x_1^2 + x_1x_2 + x_2^2) + x_3^3 - x_3x_4^2 + x_4^3 + 2x_5^3 = 0. \quad (2)$$

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Theorem 1.1 (Exact level for the cubic threefolds). *For $j \in \{1, 3\}$,*

$$\ell_{\mathbf{Q}}(X_j) = \ell_{\mathbf{C}}((X_j)_{\mathbf{C}}) = 2.$$

Equivalently, $(X_j)_{\mathbf{C}} \times \mathbf{P}^1$ is irrational, whereas $X_j \times \mathbf{P}^2$ is rational over $\mathbf{Q}$.

The following surface theorem supplies the upper bound in Theorem 1.1.

Recall that a Galois lattice M is stably permutation if $M \oplus P \simeq P'$ for some permutation Galois lattices P and P' .

Theorem 1.2 (Two-dimensional stabilization). *Let k be a field of characteristic zero and let S/k be a smooth quartic del Pezzo surface. Suppose that $S(k) \neq \emptyset$ and that $\text{Pic}(S_{\bar{k}})$ is a stably permutation $\text{Gal}(\bar{k}/k)$ -module. Then*

$$S \times \mathbf{A}^2 \quad \text{and, equivalently,} \quad S \times \mathbf{P}^2$$

are k -rational. In particular, this holds for every stably k -rational smooth quartic del Pezzo surface.

The use of universal torsors to prove stable rationality goes back to Beauville, Colliot-Thélène, Sansuc, and Swinnerton-Dyer [1]. Tschinkel and Zhang prove the rationality of universal torsors over quartic del Pezzo surfaces and classify the four minimal Galois types relevant to stable rationality. We use their Cox model and its one-apparent-double-point property, Galois classification, and generic splitting of the universal torsor [7]. The tangent-projection theorem for smooth varieties with one apparent double point is due to Ciliberto, Mella, and Russo [2, Corollary 4.2]. The projective Cox model may be singular, so we use the stronger form proved in [7, Theorem 2.4]. Our additional input is a rank-three quotient of that model.

When $S(k) \neq \emptyset$, the hypothesis on $\text{Pic}(S_{\bar{k}})$ is condition (1) of [7, Corollary 4.3]. Manin proves that stable rationality implies this condition [4, Theorem 2.2]; [7, Proposition 4.1 and Lemma 4.2] identify the remaining equivalent conditions. We prove that condition (1) implies the stronger conclusion that $S \times \mathbf{A}^2$ is rational. The stable-permutation resolution in [7, p. 13] gives a uniform bound of eleven variables. For type I_0 , a two-variable bound was already proved by Shepherd-Barron [6]; see also [7, Remark 4.4]. The quotient construction below gives the two-variable bound uniformly for all four types.

For $r \geq 0$, let $X_{j,r} \subset \mathbf{P}_{\mathbf{Q}}^{r+4}$ be obtained from equation (1) or (2) by adding $w_1^3 + \cdots + w_r^3$ to its left-hand side. Thus $X_{j,0} = X_j$.

Corollary 1.3 (Two series of cubic hypersurfaces). *For every $r \geq 0$ and $j \in \{1, 3\}$, the variety $X_{j,r} \times \mathbf{P}^2$ is rational over $\mathbf{Q}$.*

Tschinkel and Zhang supply the smoothness, fibrations, and generic-fibre Galois types in [7, Propositions 5.1–5.2]. Applying Theorem 1.2 over the two base function fields proves Corollary 1.3 and sharpens their stable-rationality conclusions to $X_{j,r} \times \mathbf{P}^2$ rational.

Corollary 1.4 (Rationality after affine-line stabilization). *For $j \in \{1, 3\}$, the variety $Y_j = X_j \times \mathbf{P}^1$ is a smooth projective fourfold over $\mathbf{Q}$ which remains nonrational over $\mathbf{C}$, while $Y_j \times \mathbf{A}^1$ is rational over $\mathbf{Q}$.*

The fourfolds Y_j answer the question posed in [7, p. 13]: find a nonrational variety Y such that $Y \times \mathbf{A}^1$ is rational. This is a birational statement, not Zariski cancellation for isomorphic affine cylinders.

Corollary 1.5 (Variation in the moduli of cubic threefolds). *On the moduli space of smooth complex cubic threefolds, $\ell_{\mathbf{C}}((X_j)_{\mathbf{C}}) = 2$ for $j \in \{1, 3\}$, whereas $\ell_{\mathbf{C}}$ is infinite at a very general point.*

Here is the proof of the surface theorem. Let Z be the projective Cox model of S . The rank-five torus T_0 acting on Z contains a saturated rank-three subtorus T_3 . Four of its weight spaces form a Galois-stable set, and their weight differences form a lattice basis. A codimension-three tangent section meets a general T_3 -orbit once, so tangent projection identifies Z/T_3 birationally with $\mathbf{P}^4$. Generic splitting of the universal torsor also gives $Z/T_3 \sim_k S \times (T_0/T_3)$. The residual torus T_0/T_3 has dimension two and is rational; the two descriptions of the quotient therefore give the rationality of $S \times \mathbf{A}^2$. Applying the argument over the two base function fields proves Corollary 1.3. In Section 2 we prove the quotient criterion, in Section 3 we construct the required Cox-model section, and in Section 4 we deduce the rationality statements.

2 A criterion for rational torus quotients

The criterion below turns a linear section into a rational quotient. Choose $r + 1$ weight spaces whose weight differences form a basis of the character lattice. Signed maximal minors then give the unique translate of a general torus orbit that lies in a suitable codimension- r linear section. If the section contains the centre of a birational tangent projection, that projection parametrizes the relevant component of the section, and hence the quotient.

Let $Z \subset \mathbf{P}(V)$ be a geometrically integral n -fold over an infinite field k . At a smooth point $p \in Z(k)$, write

$$\pi_p : Z \dashrightarrow \mathbf{P}(V/\widehat{T_p Z}) \quad (3)$$

for projection from the embedded tangent space; the hat denotes the affine cone. In the application, $Z \subset \mathbf{P}^{2n+1}$ has one apparent double point: a general point of $\mathbf{P}^{2n+1}$ lies on a unique secant line to Z . Over an algebraically closed field, its tangent projection is birational in the smooth case by [2, Corollary 4.2]. The projective Cox model need not be smooth. The form used here is the stronger [7, Theorem 2.4], which applies to a projective variety whose base change has one apparent double point and whose k -points are dense.

Theorem 2.1 (A quotient criterion). *Let a rank- r k -torus T act projectively linearly and generically freely on $Z \subset \mathbf{P}(V)$, and suppose that the generic tangent projection π_p is birational to $\mathbf{P}^n$. Choose a diagonal linear lift of the action over k^s . Its weights are determined up to a common translation. Suppose that it has nonzero weight spaces $V_{w_0}, \dots, V_{w_r}$ such that $w_1 - w_0, \dots, w_r - w_0$ is a $\mathbf{Z}$ -basis of $X^*(T_{k^s})$. Assume that the sum $W = V_{w_0} \oplus \dots \oplus V_{w_r}$ and the sum $\widehat{B}$ of all other weight spaces descend to complementary k -subspaces of V ; write $B = \mathbf{P}(\widehat{B})$.*

Suppose that there is a point $p \in Z(k)$ for which π_p is birational, together with a k -defined codimension- r linear subspace

$$\Lambda \supset \langle T_p Z, B \rangle$$

with equations $\lambda_1, \dots, \lambda_r$. Suppose there is an open $U_p \subset Z$ on which π_p is an isomorphism and a geometric point $x \in (\Lambda \cap U_p)(k^s)$ in the generically free locus such that the matrix

$$A(x)_{ij} = \lambda_i(x_j), \quad x_j \in V_{w_j}, \quad (4)$$

has rank r and all $r+1$ maximal minors nonzero. Then the Rosenlicht quotient Z/T , whose function field is $k(Z)^T$, is k -rational.

Proof. The maximal minors first produce a unique point of Λ on a general T -orbit. The hypothesis that this section meets U_p then identifies it birationally with projective space.

For a geometric point $q = \sum q_j + q_B$, where $q_j \in V_{w_j}$ and $q_B \in \widehat{B}$, put $A(q)_{ij} = \lambda_i(q_j)$ and define

$$\kappa_j(q) = (-1)^j \det A(q)_{\widehat{j}}, \quad 0 \leq j \leq r. \quad (5)$$

The equations of Λ vanish on B , so $tq \in \Lambda$ is equivalent to

$$A(q)(\chi^{w_0}(t), \dots, \chi^{w_r}(t))^t = 0.$$

The cofactor identity says that the kernel is generated by $(\kappa_0(q), \dots, \kappa_r(q))$. The basis hypothesis gives a unique $t(q) \in T(k^s)$ through

$$\chi^{w_j - w_0}(t(q)) = \frac{\kappa_j(q)}{\kappa_0(q)}, \quad 1 \leq j \leq r. \quad (6)$$

These are Laurent monomials in the displayed ratios. The construction is unchanged if all weights are translated by one character, since only their differences occur. It is also independent of the ordering of the selected weight spaces: intrinsically, $s(q) = t(q)q$ is the unique point of $Tq \cap \Lambda$. Because W , $\widehat{B}$, and Λ descend, this uniqueness makes s Galois equivariant. Thus s is a k -rational, orbit-invariant retraction on the open where the action is free and the minors do not vanish. Let C be the closure of its image. Then $C \dashrightarrow Z/T$ is birational and $C \subset \Lambda \cap Z$. Since $x \in \Lambda$, the vector $(1, \dots, 1)$ lies in the kernel of $A(x)$; uniqueness gives $s(x) = x$. Thus C meets U_p .

The image of Λ under π_p is a linear $\mathbf{P}^{n-r} \subset \mathbf{P}^n$, because Λ contains the centre of the linear projection. The irreducible variety C has dimension $n-r$, and π_p is injective on the nonempty open $C \cap U_p$. Consequently

$$\pi_p|_C : C \dashrightarrow \mathbf{P}^{n-r} \quad (7)$$

is birational. Hence Z/T is k -rational.

We record the inverse and its domain. Choose linear forms $\rho_0, \dots, \rho_{n-r}$ whose restrictions give homogeneous coordinates on the image of Λ in $\mathbf{P}(V/\widehat{T}_p Z)$. Let $U \subset Z$ be the invariant open where the action is free, all $\kappa_j(q)$ are nonzero, $s(q) \in U_p$, and $\rho_0(s(q)) \neq 0$. These conditions are invariant because s is orbit invariant and nonvanishing of the signed minors is invariant under the diagonal torus action. Then

$$\Phi : U/T \longrightarrow \mathbf{P}^{n-r}, \quad [q] \longmapsto [\rho_0(s(q)) : \dots : \rho_{n-r}(s(q))] \quad (8)$$

is the quotient parametrization. If $R_\alpha(x)$ are homogeneous equations for Z , the graph of its inverse over $\Phi(U/T)$ is the closure of the locus with $x \in C \cap U_p$ satisfying

$$\begin{aligned} R_\alpha(x) &= 0 \quad \text{for every } \alpha, \\ \lambda_i(x) &= 0 \quad 1 \leq i \leq r, \\ y_j \rho_0(x) - y_0 \rho_j(x) &= 0 \quad 1 \leq j \leq n-r. \end{aligned} \quad (9)$$

Write its unique point over y as $x(y)$ and set $\Theta(y) = [x(y)]$. The last line gives $\Phi(\Theta(y)) = y$. Conversely, $s(q) \in \Lambda \cap U_p$ and has the same tangent-projection coordinates as $x(\Phi([q]))$; injectivity of $\pi_p|_{U_p}$ gives

$$\Theta(\Phi([q])) = [s(q)] = [q] \quad \text{in } U/T.$$

Thus both composites are the identity on the stated dense opens. In the Cox application, the R_α in (9) are the twenty quadrics in [7, Section 3]. Equations (6) and (9) give the rational map and its inverse without an unrecorded finite cover. $\square$

The integral-basis condition is essential. If the weight differences span a proper finite-index sublattice, the same equations generally define a finite cover of a T -orbit rather than a rational section.

The proof is constructive once equations for Z , a tangent point, and the linear forms λ_i are given. The signed minors in (5) compute the orbit correction, and (9) recovers the inverse parametrization by elimination. Thus such a tangent section produces explicit rational maps in either direction. No estimate for the complexity of that elimination is asserted.

3 The rank-three subtorus of the projective Cox model

This section verifies the hypotheses of Theorem 2.1 for the projective Cox model. We use four facts from Tschinkel and Zhang: the model is seven-dimensional and tangent projection from a general point is birational; the scalar quotient of its rank-six Néron–Severi torus has rank five; in the minimal case the stable-permutation hypothesis leaves four possible Galois images; and a rational point on the surface gives both Zariski-dense rational points on the model and a generic splitting of the universal torsor. We construct the required rank-three subtorus and tangent section for the largest Galois type, I_3 . The other three types follow by subgroup containment after a change of marking.

Let T be the Néron–Severi torus, so $X^*(T_{\bar{k}}) = \text{Pic}(S_{\bar{k}})$. The Galois-invariant cocharacter

$$\lambda_{-K} : \mathbf{G}_m \longrightarrow T, \quad D \longmapsto (-K_S, D), \quad (10)$$

acts by simultaneous scalar multiplication on the sixteen Cox generators. Write $T_0 = T/\lambda_{-K}(\mathbf{G}_m)$, a torus of rank five. Over $\bar{k}$, the projective Cox model

$$Z = \text{Proj Cox}(S) \subset \mathbf{P}^{15}$$

is a seven-dimensional variety with one apparent double point [7, Lemma 3.2].

Use the marking of [7, Sections 3–4]. For the type I_3 Galois image, the sublattice

$$N_3 = \mathbf{Z}e_3 \oplus \mathbf{Z}e_4 \oplus \mathbf{Z}e_5 \subset X_*(T_{0,\bar{k}}) \quad (11)$$

is saturated and Galois stable. The two displayed generators of the type- I_3 Galois image act on this basis by

$$\begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

Let $T_3 \subset T_0$ be the corresponding rank-three subtorus. Four groups of Cox generators span weight spaces with the following projective T_3 -weights:

weight	Cox generators
$(0, 1, 1)$	L_{13}, L_{23}, L_{35}
$(1, 0, 1)$	L_{14}, L_{24}, L_{45}
$(1, 1, 0)$	E_1, E_2, E_5
$(1, 1, 1)$	L_{12}, L_{15}, L_{25}

These four weight spaces form a Galois-stable set, but they are not individually Galois stable. On their weights, the first displayed generator acts by

$$(011 \ 111)(101 \ 110),$$

and the second by (110 111), fixing 011 and 101. These are affine actions on projective weights: a different diagonal lift translates all four weights by a common character. Every other Cox generator lies in the Galois-stable subspace

$$B = \mathbf{P}\langle E_3, E_4, L_{34}, Q \rangle. \quad (12)$$

The determinant of the three weight differences from $(0, 1, 1)$ is one. Consequently the direct sum of the four selected weight spaces and the affine cone over B are complementary subspaces defined over the ground field, as required in Theorem 2.1. On the open where the four selected weight-space components are nonzero, a projective stabilizer fixes their three ratios. The three weight differences form a basis, so that stabilizer is trivial. In particular, the T_3 -action is generically free.

Remark 3.1 (Saturation and the projective kernel). In the type- I_1 description, the three visible sign cocharacters in the last three coordinates are

$$(0, 1, 1), \quad (1, 0, 1), \quad (1, 1, 0).$$

They span the even-sum sublattice of $\mathbf{Z}^3$, of index two. Its saturation is $\mathbf{Z}^3$, obtained by adjoining

$$(1, 1, 1) = \frac{1}{2}((0, 1, 1) + (1, 0, 1) + (1, 1, 0)).$$

The map from the three parameter copies of $\mathbf{G}_m$ to their image in the projective torus has kernel μ_2 . The degree two obtained from the unsaturated parameters counts this kernel; it is not the degree of the Rosenlicht quotient. Saturation gives the integral-basis condition in Theorem 2.1.

Proposition 3.2 (A tangent section for the four Galois types). *Over an algebraically closed field, let Z be the projective Cox model of a smooth quartic del Pezzo surface. There exist a general tangent point p , a point x , and a codimension-three linear subspace $\Lambda \supset \langle T_p Z, B \rangle$ such that $x \in \Lambda \cap U_p$ and the maximal-minor condition of Theorem 2.1 holds at x for T_3 . If Z is the projective Cox model of a smooth quartic del Pezzo surface S/k with $S(k) \neq \emptyset$, and the Galois image has type I_0, I_1, I_2 , or I_3 , such a triple (p, x, Λ) exists over the ground field.*

Proof. Use the standard blowup points

$$(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1), (1 : 1 : 1), (1 : a : b), \quad \Delta = ab(a-1)(b-1)(a-b) \neq 0. \quad (13)$$

For a tangent point with auxiliary coordinates $z = (z_1, z_2, z_3)$, the space of tangent hyperplanes containing B has dimension four. Evaluation on the four selected weight spaces gives a 4-by-4 matrix. The following exact choices are used; $e = (e_1, \dots, e_5)$ and z' specify the point x used to evaluate the matrix.

i	z	e	z'
1	(1, 3, 7)	(2, 3, 5, 7, 11)	(2, 4, 9)
2	(2, 5, 11)	(3, 4, 7, 13, 17)	(1, 6, 10)
3	(3, 8, 13)	(5, 7, 11, 17, 19)	(2, 9, 15)
4	(4, 9, 17)	(2, 5, 11, 19, 23)	(3, 10, 18)

Here e gives the five coordinates $E_1, \dots, E_5$. From e and z' , the public generator constructs the remaining coordinates L_{ij} and Q and verifies directly that the resulting point satisfies all twenty Cox quadrics.

Let D_i be the determinant of this evaluation matrix, and let M_i be the fixed 8-by-8 Jacobian minor at the same tangent point. After discarding nonzero rational factors, their numerators are recorded in Appendix A. Choice i works precisely on the open $D_i M_i \neq 0$. The appendix proves, by eight localized cases and one explicit Bézout identity, that

$$\bigcup_{i=1}^4 \{D_i M_i \neq 0\} = \{\Delta \neq 0\}. \quad (14)$$

Let $\mathcal{E}_{\text{ev}} \subset Z_{\text{sm}} \times Z$ be the open locus where the evaluation map is an isomorphism and the T_3 -action is free at x . The four choices above show that this open is nonempty for every split Cox model attached to a smooth quartic del Pezzo surface.

We now impose the tangent-projection condition uniformly in p . Let $G \subset Z_{\text{sm}}$ be the nonempty open over which the tangent projection is birational. The affine tangent spaces form a subbundle $\widehat{\mathcal{T}} \subset V \otimes \mathcal{O}_G$. Put

$$\mathcal{Q} = (V \otimes \mathcal{O}_G) / \widehat{\mathcal{T}}.$$

The tangent projections are the fibres of the rational G -map

$$\Pi : G \times Z \dashrightarrow \mathbf{P}_G(\mathcal{Q}), \quad (p, x) \longmapsto (p, \pi_p(x)).$$

The induced map on the generic fibre of G is birational, so Π restricts to an isomorphism between dense open subsets of its source and target. Denote the source open by $\mathcal{U}$. Thus $(p, x) \in \mathcal{U}$ is a single open condition on the pair, and its fibre over p is an open U_p on which π_p is an isomorphism. Since $G \times Z$ is geometrically integral, the open $\mathcal{U} \cap (\mathcal{E}_{\text{ev}}|_{G \times Z})$ is nonempty.

Choose (p, x) in this intersection. Then p is general for tangent projection, $x \in U_p$, the evaluation map is an isomorphism at x , and the T_3 -action is free at x .

Let H_p be the four-dimensional space of tangent hyperplanes containing B . Evaluation at x on the four selected weight-space components defines an isomorphism $H_p \simeq k^4$. Hyperplanes in H_p that vanish at x form a three-dimensional subspace; choose a basis $\lambda_1, \lambda_2, \lambda_3$ and let Λ be their common zero locus. Then $x \in \Lambda \cap U_p$. Under the evaluation isomorphism the three rows span the hyperplane $u_0 + u_1 + u_2 + u_3 = 0$, so their four maximal minors are all nonzero. Thus the point satisfying the minor condition is the same point that lies in the tangent-projection isomorphism open.

It remains to obtain p , x , and Λ over the ground field. Fix a twist of one of the four types. Because T_3 , B , the set of four selected weight spaces, and the universal tangent-projection map are defined over k , the intersection $\mathcal{U} \cap (\mathcal{E}_{\text{ev}}|_{G \times Z})$ is a nonempty k -open in $G \times Z$. Its projection to G is dense constructible and therefore contains a nonempty open G_0 .

Corollary 3.5 of [7] supplies the universal torsor whose fibre over a fixed point of $S(k)$ is trivial. Its k -points are Zariski dense by [7, Lemma 2.1], and the same holds in the projective Cox model. Hence G_0 contains a point $p \in Z(k)$. The fibre $(\mathcal{U} \cap (\mathcal{E}_{\text{ev}}|_{G \times Z}))_p$ is a nonempty k -open in Z , so Zariski density supplies a point $x \in Z(k)$ in that fibre. The kernel of evaluation at x is a three-dimensional k -space, so a k -basis gives Λ . We do not claim that any one of the four displayed coordinate choices descends.

For type I_3 , stability of N_3 , B , and the set of four selected Cox weight spaces is checked on the two displayed generators. Up to conjugacy in $W(D_5)$, the groups of types I_0, I_1, I_2 are subgroups of the group of type I_3 [7, Lemma 4.2]. Restriction and a change of marking preserve the same data. This exhausts the four Galois images allowed by the hereditary H^1 -condition; it is not a classification of arbitrary subgroups of $W(D_5)$. $\square$

4 Proofs of the rationality statements

Proof of Theorem 1.2. We first treat the minimal case by taking a quotient of the projective Cox model. An equivariant generic trivialization then identifies that quotient with the product of S and a two-dimensional torus. The nonminimal case reduces to the classical rationality theorem in degree at least five.

Suppose first that S is k -minimal. Since a stably permutation module has vanishing H^1 on every open subgroup, Proposition 4.1 and Lemma 4.2 of [7] place the Galois image in one of the four types used in Proposition 3.2. Lemma 3.2 and Theorem 2.4 of [7], together with Theorem 2.1, give

$$Z/T_3 \sim_k \mathbf{P}^4. \quad (15)$$

By [7, Corollary 3.5], let $\mathcal{T} \rightarrow S$ be a universal torsor under the Néron–Severi torus T , chosen to have a k -point above a fixed point of $S(k)$. The stable-permutation identity for $X^*(T_{\bar{k}})$ expresses T , after multiplication by a quasi-trivial torus, as another quasi-trivial torus. Hilbert’s Theorem 90 therefore gives

$$H^1(F, T) = 0 \quad \text{for every field extension } F/k \quad (16)$$

by [7, Remark 2.2]. Thus the generic fibre over $k(S)$ has a point. Equivalently, the torsor has a rational section σ . Translation by σ defines the T -equivariant birational map

$$S \times T \dashrightarrow \mathcal{T}, \quad (s, t) \mapsto \sigma(s)t. \quad (17)$$

The split scalar subtorus in (10) acts as common multiplication on every Cox coordinate. The universal torsor is a dense open in the affine Cox cone, and its scalar quotient is a dense open in Z . Quotienting (17) by that scalar therefore gives a T_0 -equivariant birational map $S \times T_0 \dashrightarrow Z$. Since $T_3 \subset T_0$ is a k -subtorus, Rosenlicht quotients give

$$Z/T_3 \sim_k S \times (T_0/T_3). \quad (18)$$

The quotient T_0/T_3 is a two-dimensional k -torus, hence is k -rational by [9]. Equations (15) and (18) prove that $S \times \mathbf{A}^2$ is k -rational.

If S is not k -minimal, successively contracting k -defined unions of pairwise disjoint exceptional curves gives a del Pezzo surface S' of degree at least five. The image of a k -point of S is a k -point of S' , so S' , and therefore S , is k -rational by [8, Theorem 2.1].

Finally, if S is stably k -rational, a nonempty open of a projective space is birational to a nonempty open of $S \times \mathbf{P}^m$. Since k is infinite, this supplies a k -point on S . Manin’s stable birational Picard-module obstruction implies that $\text{Pic}(S_{\bar{k}})$ is stably permutation. In the form used here, this consequence of [4, Theorem 2.2] is recalled immediately before [7, Proposition 4.1] and included in [7, Corollary 4.3]. This proves the last assertion. $\square$

Proof of Corollary 1.3. The argument takes place entirely over the function field of the projection base. This avoids any specialization or fibrewise rationality claim.

For $X = X_{j,r}$, Tschinkel and Zhang project from a plane containing a distinguished line. The base is $\mathbf{P}_{\mathbf{Q}}^{r+1}$, with function field $K = \mathbf{Q}(t_1, \dots, t_{r+1})$, and the generic cubic-surface fibre is K -birational, after contracting the distinguished line, to a smooth quartic del Pezzo surface S/K with a K -point. Their calculation identifies the Galois image as type I_1 for $j = 1$ and type I_3 for $j = 3$ [7, Propositions 5.1–5.2]. Lemma 4.2 of [7] then shows that $\text{Pic}(S_{\bar{K}})$ is stably permutation. Theorem 1.2 gives

$$K(S)(u, v) \simeq K(z_1, z_2, z_3, z_4).$$

Since $\mathbf{Q}(X) = K(S)$ as extensions of K ,

$$\mathbf{Q}(X)(u, v) \simeq \mathbf{Q}(t_1, \dots, t_{r+1}, z_1, z_2, z_3, z_4)$$

is purely transcendental. This is rationality of $X \times \mathbf{A}^2$, equivalently of $X \times \mathbf{P}^2$. This is a function-field argument; no specialization assertion is used. $\square$

Proof of Theorem 1.1. The one-stabilization theorem proves that $X \times \mathbf{P}^1$ is irrational for every smooth complex cubic threefold X [5, Theorem 1.1]. It applies to $(X_j)_{\mathbf{C}}$. Therefore neither X_j nor $X_j \times \mathbf{P}^1$ can be rational over $\mathbf{Q}$, because rationality survives extension of the ground field. Corollary 1.3 gives rationality at $m = 2$ over $\mathbf{Q}$, and hence also over $\mathbf{C}$. This proves both equalities of stabilization levels. $\square$

Proof of Corollary 1.4. Theorem 1.1 shows that $Y_j = X_j \times \mathbf{P}^1$ is nonrational, even after base change to $\mathbf{C}$. On the other hand,

$$\mathbf{Q}(Y_j \times \mathbf{A}^1) = \mathbf{Q}(X_j)(u, t)$$

is purely transcendental by Corollary 1.3. Equivalently, $\mathbf{P}^1 \times \mathbf{A}^1$ is birational to $\mathbf{A}^2$. Hence $Y_j \times \mathbf{A}^1$ is rational. $\square$

Proof of Corollary 1.5. Theorem 1.1 gives the asserted value for X_1 and X_3 . Engel, de Gaay Fortman, and Schreieder prove that a very general smooth complex cubic threefold does not admit a decomposition of the diagonal and hence is not stably rational [3, Corollary 1.4]. For such a threefold no finite m makes $X \times \mathbf{P}^m$ rational, so $\ell_{\mathbf{C}}(X) = \infty$. $\square$

A Exact tangent-section certificate

This appendix proves that the four opens used in Proposition 3.2 cover the smooth parameter space. If the first three products $D_i M_i$ vanish, choosing one vanishing factor from each product gives eight cases. Exact Gröbner computations show that six cases are empty. In each of the other two, the displayed evaluations and Bézout identity show that $D_4 M_4 \neq 0$. The public certificate retains, for each empty case, a machine-readable identity expressing a nonzero constant in the corresponding localized ideal.

After discarding nonzero rational factors, the four evaluation determinants are

$$\begin{aligned} D_1 &= (8a - 27)^2 (3968a - 1349b - 2619), \\ D_2 &= 568453977a^3 - 373279764a^2b - 1658788404a^2 \\ &\quad + 59068867ab^2 + 976763788ab + 925668016a \\ &\quad - 128537024b^2 - 359163728b - 10185728, \\ D_3 &= 1643918400a^3 - 1105797420a^2b - 7261914240a^2 \\ &\quad + 154718524ab^2 + 3804690316ab + 9944961600a \\ &\quad - 282623593b^2 - 3443769343b - 3454184244, \\ D_4 &= 35068545a^3 + 2864538a^2b - 236376549a^2 \\ &\quad - 5350827ab^2 + 24417420ab + 469658574a \\ &\quad + 14536855b^2 - 90750394b - 214068162. \end{aligned}$$

and the corresponding smoothness minors are

$$\begin{aligned} M_1 &= -4(a-1)(b-7)(3a-b-2)(2ab+7a-9b), \\ M_2 &= -9(a-1)(2b-11)(3a-b-2)(4ab+11a-15b), \\ M_3 &= -25(a-1)(3b-13)(2a-b-1)(3ab+13a-16b), \\ M_4 &= -(a-1)(4b-17)(13a-5b-8)(32ab+85a-117b). \end{aligned}$$

To prove (14), suppose first that $D_i M_i = 0$ for $i = 1, 2, 3$. Choosing one factor from each product gives eight cases. A lexicographic Gröbner calculation in $\mathbf{Q}[a, b, h]/(h\Delta - 1)$ gives

chosen equations	localized zero locus
$D_1 D_2 D_3, D_1 D_2 M_3, D_1 M_2 D_3, D_1 M_2 M_3$	empty
$M_1 D_2 D_3, M_1 D_2 M_3$	empty
$M_1 M_2 D_3$	$3a - b - 2 = Q_0(b) = 0$
$M_1 M_2 M_3$	$3a - b - 2 = (3b - 26)(3b - 13) = 0$

where

$$Q_0(b) = 31223016b^2 - 435944529b + 1306078948.$$

On the line $3a - b - 2 = 0$, the primitive numerator of $D_4 M_4$ is

$$\begin{aligned} &-(b-1)^4(4b-17)(16b-85)(83246b^2 - 872181b + 2185995), \\ Q_4(b) &= 83246b^2 - 872181b + 2185995. \end{aligned}$$

The evaluations

$$\begin{aligned} Q_0(17/4) &= \frac{69121705}{4}, & Q_0(85/16) &= -\frac{4117757269}{32}, \\ Q_4(26/3) &= \frac{7918133}{9}, & Q_4(13/3) &= -\frac{272530}{9} \end{aligned}$$

exclude the four linear roots. Finally, the two quadratics are coprime by the identity

$$\begin{aligned} &(2265746679974131615 - 377042650728395274b)Q_0(b) \\ &+ (141417109727495582904b - 1342668830289072756147)Q_4(b) \\ &= 24176690547344887359179755. \end{aligned}$$

Thus $D_4 M_4 \neq 0$ in both remaining cases. This proves that the four opens $D_i M_i \neq 0$ cover $\Delta \neq 0$.

B Verification and scope

Using exact integer and rational arithmetic, we compute the cocharacter saturation, type- I_3 actions, and Cox weights; verify that the selected weight differences form a lattice basis; and prove that four explicit opens cover the smooth parameter locus. The public generator starts from the two type- I_3 character matrices and a transcription of the twenty Cox quadrics in [7, Section 3]. It reconstructs the saturated rank-three lattice, the residual rank-two quotient, the tangent Jacobian, all four pairs (D_i, M_i) , the eight localized cases, and the Bézout identity above with SymPy 1.14.0. The certificate retains exact Nullstellensatz identities for the six empty cases; both public programs verify those identities directly. The generator also derives the linear section defined by the kernel of evaluation at x and renders

every computation-derived matrix, weight-space table, parameter row, determinant, minor, and Bézout coefficient printed in the manuscript into a checked TeX artifact. A second checker independently verifies the resulting cover, linear section, and quotient minors. We check the transcription of the source equations, the quotient criterion, equivariant generic trivialization, incidence descent, and function-field argument in the text; the programs do not address them.

No Lean development formalizes Theorem 1.1, Theorem 1.2, Theorem 2.1, Proposition 3.2, or the stated corollaries. The formal-coverage annotations therefore say **absent**.

We import Tschinkel–Zhang’s universal-torsor geometry and their classification of the four hereditary- H^1 Galois images. We construct the rational rank-three quotient and its descended tangent section. This construction lowers the number of stabilizing variables to two; no classification beyond those four Galois types is needed here.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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